APPLE QUALITY PREDICTION USING MACHINE LEARNING

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Team_Name:- Data Warriors

PROBLEM STATEMENT

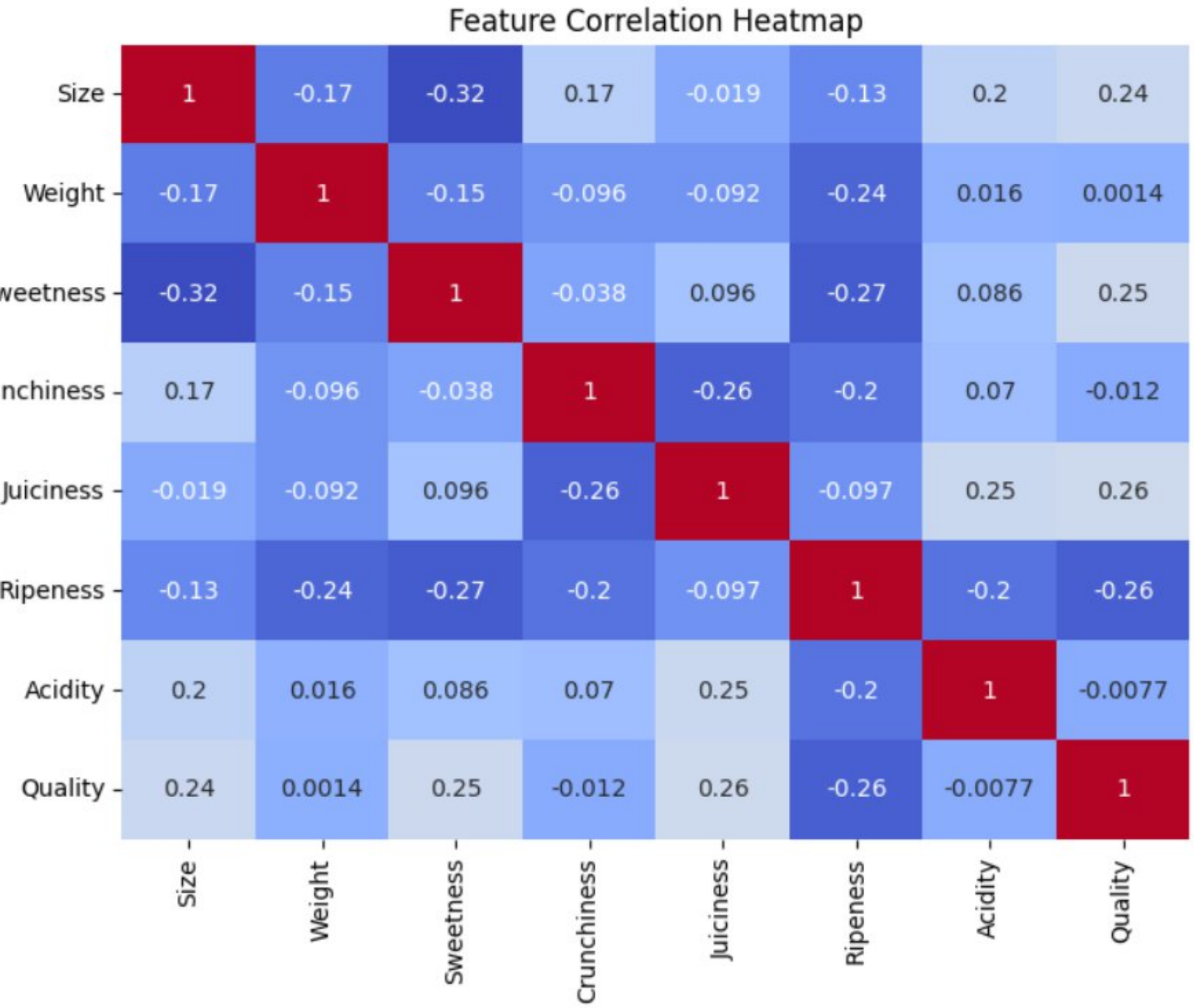
- Manual fruit grading is inconsistent
- Time-consuming and subjective
- Need **automated ML solution**

DATASET OVERVIEW

4001 samples

- 7 features
- Target: Good / Bad

	A_id	Size	Weight	Sweetness	Crunchiness	Juiciness	Ripeness	Acidity	Quality
0	0.0	-3.970049	-2.512336	5.346330	-1.012009	1.844900	0.329840	-0.49159	good
1	1.0	-1.195217	-2.839257	3.664059	1.588232	0.853286	0.867530	-0.722809	good
2	2.0	-0.292024	-1.351282	-1.738429	-0.342616	2.838636	-0.038033	2.621636	bad
3	3.0	-0.657196	-2.271627	1.324874	-0.097875	3.637970	-3.413761	0.790723	good
4	4.0	1.364217	-1.296612	-0.384658	-0.553006	3.030874	-1.303849	0.501984	good



EXPLORATORY DATA ANALYSIS

- **Analysis Performed**
- Feature distribution analysis
- Correlation analysis
- Data visualization
- **Key Observations**
- Sweetness and Juiciness vary significantly
- Ripeness shows high variation
- Dataset is balanced

KEY INSIGHTS

- **Core Insights**
- **Juiciness is the strongest indicator of apple quality**
- **Higher sweetness significantly increases quality**
- **Over-ripeness negatively impacts apple quality**

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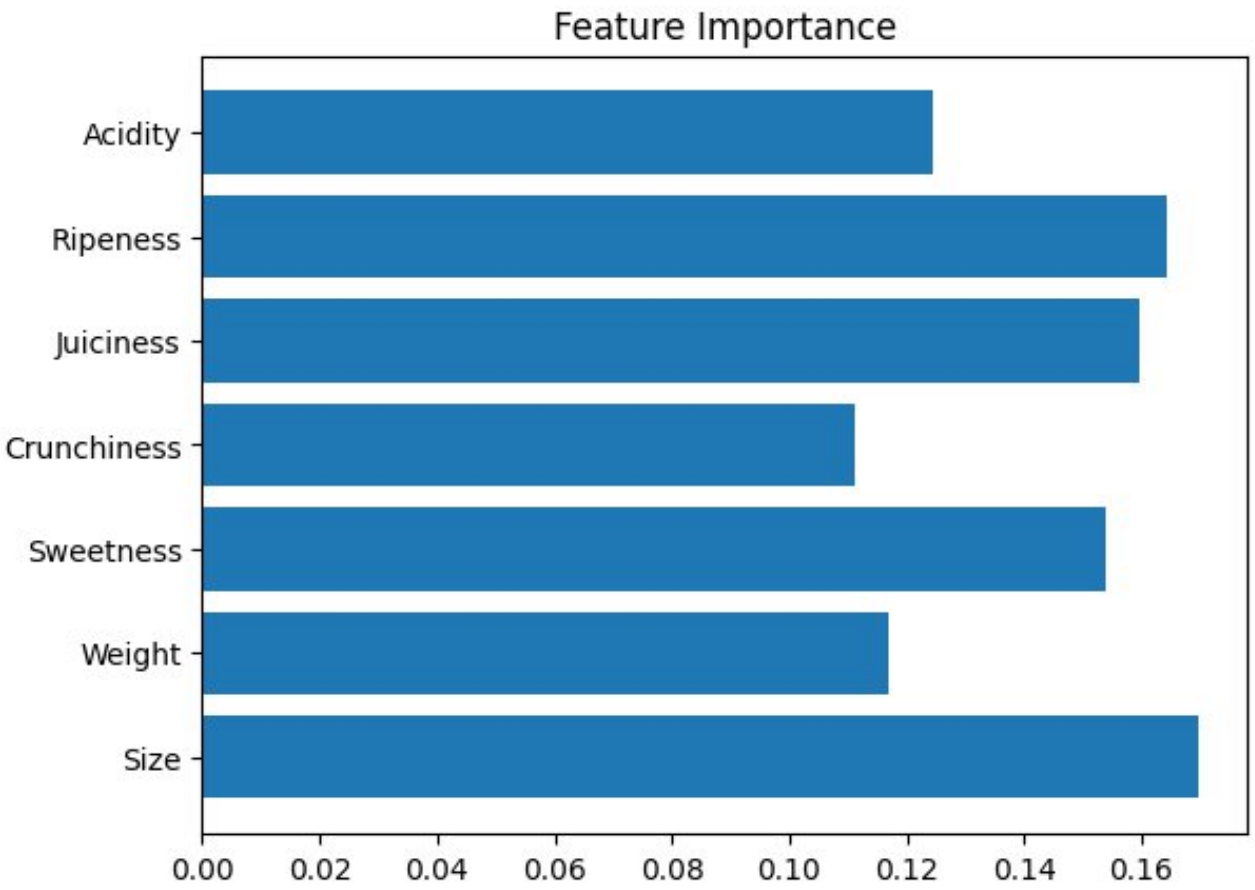
MACHINE LEARNING & RESULTS



Models Used

- Logistic Regression
- Random Forest
- Support Vector Machine
- **Best Model**

☞ Random Forest (~90% Accuracy)

- **Key Features**
- Juiciness
- Sweetness
- Ripeness

CONCLUSION

- **Conclusion**
- Machine learning can automate apple quality prediction
- Data-driven approach improves consistency
- **Applications**
- Agriculture
- Food Industry
- Supply Chain

Thank You